

Streamlining Ancestry: A Lightweight Document Processing Pipeline for Historical Document Data Extraction

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This project illustrates an innovative approach to solving the bottlenecks inherent in manual genealogical data extraction. Historical records, such as birth, marriage, and death certificates hold vital, cultural and legal value, yet manual transcription is an arduous, time-consuming process prone to human error. Our research introduces a lightweight, scalable system leveraging Artificial Intelligence through Object detection and Optical Character Recognition designed to automate the extraction of information from diverse document formats across various time periods and regions.

Our goals are to establish a lightweight system utilizing a deep learning pipeline that meets the following requirements.

- Accuracy: Identifies one document format with a high level of accuracy (>96%)
- Simple: requires 15-20 pieces of labeled data to train
- Replicable: easily replicable to quickly create a model that will recognize a new document format

The core of this project is the development of a rapid-deployment pipeline that minimizes the labor required to build document-specific recognition systems. By utilizing a YOLOv8 (You Only Look Once) object detection network, the system identifies and extracts handwritten regions of interest from scanned documents. These segments are then processed by a TrOCR (Transformer-based Optical Character Recognition) model to convert the images to text, ultimately outputting structured data into a CSV format. A key innovation of this work is its

efficiency; each specific document type requires only 10–20 labeled images for training, making it a highly accessible tool for smaller archives and large-scale databases

References

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